Welcome to your next lesson: String membership with 'in' in conditions.

I'm preparing your personalized lesson based on what you've learned, what's coming up next, and the best way to strengthen your skills.

Step by step, you're moving forward.

Let's start with something real-world and simple ?

Imagine you have a text message:

Python
```
message = "URGENT: server is down"
```

You can ask Python a yes/no question like this:

Python
```
"URGENT" in message
```

That expression returns either **True** or **False** depending on whether "URGENT" appears anywhere inside **message**.

So you can use it in an if:

Python
```
if "URGENT" in message:
    print("Handle this right now!")
```

Does this idea make sense so far: "<piece of text>" in "<bigger text>" gives you **True** if it's inside, otherwise **False**?

Yes, it makes sense

What do you think this code does? Take your time and pick an answer below ?

What will this print?

```
message = "Meeting at 3pm"

if "URGENT" in message:
    print("High priority")
```

A.

High priority

B.

Meeting at 3pm

C.

It will print nothing

D.

It will cause an error

Nice work, Ron — you nailed it ?

Your answer "It will print nothing" is exactly right, because:

- "URGENT" in **message** checks if the substring "URGENT" is inside "Meeting at 3pm".
- It's not there, so the condition is **False**.

- When an `if` condition is **False**, the code inside the `if` block is skipped, so nothing is printed.
  Next step: I want to show you the opposite check using `not in` (like “if 'URGENT' is not in the message”).
  Ready for that? ✅ **?**

  Yes, I'm ready

  Great ?
  Here’s the mirror image of what you just saw, using `not in`:

```python
message = "Meeting at 3pm"

if "URGENT" not in message:
    print("This is not urgent.")
```

  Step by step:

1. Python checks: is `"URGENT"` not inside `"Meeting at 3pm"`?
2. That’s True, because the word isn’t there.
3. So the `if` body runs, and it prints: `This is not urgent.`

  So:

  `in` → True when the small text is inside

  `not in` → True when the small text is not inside

  Does this flip between `in` and `not in` feel clear so far?

  Yes, it's clear

  What do you think this one does? Pick the answer that feels right ?
  What will this code do?

```python
message = "Lunch at noon"

if "URGENT" not in message:
    print("All good, no rush.")
```

  A.
  Print: All good, no rush.
  B.
  Print: URGENT
  C.
  Print nothing
  D.
  Cause an error, because of `not in`
  You’re on a roll, Ron ?
  Your answer “Print: All good, no rush.” is exactly right.

Quick breakdown:

- "URGENT" **not** in **message** asks: Is "URGENT" missing from "Lunch at noon"?
- That's True (the word isn't there).
- Since the if condition is True, Python runs the block and prints:
  All good, no rush.

So you've now seen both directions:

- **if** "URGENT" in **message** → runs when it is there
- **if** "URGENT" **not** in **message** → runs when it's not there

Next, I'd like to plug this into a slightly bigger decision with if / **elif** / **else** (like checking for "URGENT", then "FYI", otherwise "normal message").

Want to try that next?

Yes, let's do it

Awesome ?

Let's build a tiny "message classifier" using if, **elif**, **else** and in:

Python

```
message = "FYI: server will restart tonight"

if "URGENT" in message:
    print("High priority")
elif "FYI" in message:
    print("Just for your information")
else:
    print("Normal message")
```

Step by step:

1. Python first checks: "URGENT" in **message**
2. If that's False, it moves to **elif** "FYI" **in** message
3. If both are False, it runs the **else** block.

In this example, "FYI" is in the message, so it should print:

Just for your information

Make sense how in fits inside if / **elif** like this?

If yes, I'll have you predict the output of a similar one next.

Yes, that makes sense

What do you think this program will print? Pick one option below ?

Given this code, what will be printed?

```
message = "Reminder: pay bills"

if "URGENT" in message:
```

```
    print("High priority")
elif "Reminder" in message:
    print("Scheduled task")
else:
    print("Other message")
```

A.

High priority

B.

Scheduled task

C.

Other message

D.

Nothing will be printed

E.

It will cause an error

You're doing great with these, Ron ?

You picked "Scheduled task", which is exactly right.

Why it works:

- First check: `"URGENT" in message` → `"URGENT"` is not in `"Reminder: pay bills"` → condition is False → skip that block.
- Second check: `"Reminder" in message` → `"Reminder"` is in the text → condition is True → it runs: `print("Scheduled task")`
- Because one `elif` matched, Python never reaches the `else`.

So you just used `in` inside an `if / elif / else` chain correctly ?

Want to try writing a small one yourself in the code editor?

I can give you a clear, tiny task, and you'll code it.

Yes, I'd like to try

Nice, let's do this step by step ?□˚ﾒ□

Here's your task to do in the code editor (not in chat):

1. Create a variable called `message` with some text, like:
   `message = "URGENT: password reset needed"`
2. Write an `if / elif / else` that does this:

- If `"URGENT"` is in `message`, `print("High priority")`
- `elif "FYI"` is in `message`, `print("Informational")`
- `else`, `print("Regular message")`

So the structure should look like:

Python

```
message = "..."  # your text
```

```
if "URGENT" in message:
    print("High priority")
elif "FYI" in message:
    print("Informational")
else:
    print("Regular message")
```

Go ahead and:

- Open the code editor,
- Type this program,
- Pick any message text you like,
- Hit Run to test, then Submit code when you're happy with it.

I'll check it and we'll tweak it if needed ?

```
Python
message = "URGENT: password reset needed"
if "URGENT" in message:
    print("High Priority")
elif "FYI" in message:
    print("Informational")
else:
    print("Reguar message")
```

Nice job, Ron — that's exactly the structure I was hoping for ?

Your code:

```
Python
message = "URGENT: password reset needed"
if "URGENT" in message:
    print("High Priority")
elif "FYI" in message:
    print("Informational")
else:
    print("Reguar message")
```

Why it works:

- "URGENT" is in the message, so the first `if` is `True`.
- Python runs `print("High Priority")`, which matches the output you saw.
- Because the `if` was True, it skips the `elif` and `else`.

Only tiny thing: "Reguar message" has a small spelling typo (missing the second “l” in “Regular”), but that doesn’t affect the logic at all.

You’ve now:

- Used in and not in on strings ✅
- Put membership checks inside if / elif / else to drive decisions ✅

You’ve hit all the goals for this lesson ?

Are you ready to wrap up this lesson and move on to the next one, or do you want one more quick practice with in / not in first?

I'm ready for the next lesson

Awesome work today, Ron ?

Quick recap of what you just mastered:

- You used "text" in bigger_text to check if a substring is present.
- You flipped the logic with "text" not in bigger_text to handle the “missing” case.
- You plugged those checks into full if / elif / else chains to make real decisions based on message content (like URGENT vs FYI vs regular).

You’re thinking like a programmer: letting Python answer questions about text and acting on the result. That’s a powerful pattern you’ll reuse a lot. ?

You can continue to the next lesson from the Learn tab whenever you’re ready.

Before you go on: anything still fuzzy about in / not in, or are you feeling solid?

Thank you for your feedback!